Convex Lens Problems: Worked Solutions

Formula: 1/u + 1/v = 1/f

Sign convention: real is positive. Distances for real objects and real images are positive. A negative image distance means the image is virtual.

Question 1

An object is 30 cm from a converging lens and the image is formed 60 cm from the lens. Calculate the focal length of the lens.

Step 1: Given. u = 30 cm, v = 60 cm, f = ?

Step 2: Formula. 1/u + 1/v = 1/f

Step 3: Substitute. 1/f = 1/30 + 1/60

Step 4: Common denominator. 1/f = 2/60 + 1/60 = 3/60 = 1/20

Step 5: Answer. f = **20 cm**

Question 2

An object is 20 cm from a convex lens, of focal length 60 cm. Calculate the distance from the image to the lens. Is the image real or virtual?

Step 1: Given. u = 20 cm, f = 60 cm, v = ?

Step 2: Formula. 1/v = 1/f − 1/u

Step 3: Substitute. 1/v = 1/60 − 1/20

Step 4: Common denominator. 1/v = 1/60 − 3/60 = −2/60 = −1/30

Step 5: Answer. v = −30 cm. The image is **30 cm** from the lens and is **virtual**.

The negative sign shows the image is on the same side of the lens as the object. The object is inside the focal length, so the lens acts as a magnifying glass.

Question 3

An object is 15 cm from a convex lens of focal length 10 cm. Calculate the distance from the image to the lens. Is the image real or virtual?

Step 1: Given. u = 15 cm, f = 10 cm, v = ?

Step 2: Formula. 1/v = 1/f − 1/u

Step 3: Substitute. 1/v = 1/10 − 1/15

Step 4: Common denominator. 1/v = 3/30 − 2/30 = 1/30

Step 5: Answer. v = +30 cm. The image is **30 cm** from the lens and is **real**.

The positive sign shows the image is on the opposite side of the lens from the object. The object is between f and 2f, so the image forms beyond 2f.

Question 4

An object is 40 cm from a convex lens and the image is formed 60 cm from the lens. Calculate the focal length of the lens.

Step 1: Given. u = 40 cm, v = 60 cm, f = ?

Step 2: Formula. 1/u + 1/v = 1/f

Step 3: Substitute. 1/f = 1/40 + 1/60

Step 4: Common denominator. 1/f = 3/120 + 2/120 = 5/120 = 1/24

Step 5: Answer. f = **24 cm**